Arjun Dindigal

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Technical Skills

Mechanical Design & Analysis: AutoCAD, DFMA, FEA, GD&T, ONshape, Siemens NX, SolidWorks
Manufacturing: CNC machining (mill/lathe), composites, FDM/SLA 3D printing, rapid prototyping
Programming/Firmware: C++, Python, MATLAB, Arduino, ROS, RTOS, STM32, Git

Experience

Mechanical Engineer — Humanoid Team May 2026 – Present
WATonomous *Waterloo, ON*

- Designing humanoid arm and forearm assemblies in **Onshape** with **GD&T** tolerancing, performing **ANSYS FEA** to validate structural integrity and drive design iterations prior to fabrication.
- Selecting actuators and motors for upper-limb degrees of freedom based on torque, backdrivability, and form factor constraints, and fabricated prototypes via **3D printing** to validate fit and function.
- Integrating arm and forearm subsystems across mechanical, electrical, and **ROS 2** software interfaces, executing functional testing and resolving design issues through structured validation loops.

Mechanical Design Engineer Jan. 2026 – May 2026
Linamar Corporation *Guelph, ON*

- Designed and validated **10+** production fixtures and mechanical components using **SolidWorks**, applying **GD&T** and **DFMA** to ensure manufacturability and precision.
- Reduced cycle times by **20–40 seconds** per operation by redesigning fixture setups and improving ergonomics in collaboration with machinists and operators.
- Reverse engineered failed electromechanical components from automated systems, producing CAD models and fabricated replacements to restore uptime and improve system reliability.
- Developed rapid **FDM/SLA** prototypes for iterative design testing, accelerating iteration cycles and reducing production delays.
- Improved manufacturing quality by reducing defective parts by **10%+** through fixture redesigns, cycle time studies, and optimized part handling.
- Took designs from **concept to CAD to fabrication to validation** with shop floor teams to deploy solutions.

Software Development Intern May 2025 – Aug. 2025
Pratyin Infotech Consulting Pvt. Ltd *Toronto, ON*

- Developed Java + FastAPI integration to synchronize **5,000+** invoices with **99.7% accuracy**, improving reliability of internal financial systems.
- Automated **Python** data pipelines to process **200,000** records monthly, reducing manual processing time by **20%** and increasing team efficiency.
- Delivered **2** production-ready features in **2 months** within an Agile workflow, with **zero critical bugs** deployed.

Projects

Regress Cart | *Siemens NX, SolidWorks, DFMA, GD&T* **Feb. 2026 – Mar. 2026**

- Reverse engineered an industrial production cart for setup technicians, reducing part count and eliminating unnecessary fasteners through **DFMA** and ergonomic optimization.
- Led design-to-manufacturing workflow by generating **GD&T drawings** and coordinating waterjet fabrication with machinists.
- Assembled and validated a **100+ kg** load-bearing mechanical system, improving setup efficiency and reducing cycle time.

Autonomous Retrieval VEX Bot | *C++, PID Control, Sensor Fusion, CAD* **Oct. 2025 – Dec. 2025**

- Developed **PID-based** navigation and sensor fusion algorithms in **C++**, achieving **98%** turn accuracy and **92%** object retrieval success.
- Designed and manufactured **8+** custom components, reducing system weight by **15%** while maintaining structural integrity.
- Optimized control logic for consistent autonomous performance, completing runs in **40–45 seconds** with **±1 second** deviation.

Education

University of Waterloo Sep. 2025 – Apr. 2030
Candidate for B.A.Sc. in Mechatronics Engineering (Excellent Standing) *Waterloo, ON*